

Credibility of Fluid–Structure Interaction Models for Aortic Valve Leaflets: Multi-Model, Multi-Mechanism Assessment Focused on Coaptation and Stress

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Abstract

Purpose: We assess the credibility of three fluid–structure interaction (FSI) computational models for bioprosthetic trileaflet aortic valves against the ASME V&V 40 risk-informed framework. The analysis is conducted per model and per mechanism of interest with explicit factor-by-factor scoring on a 0–4 Credibility Index. **Scope:** Mechanisms cover diastolic coaptation integrity, systolic leaflet stress at peak flow, and cycle-to-cycle stress range at commissures. **Models include:** (A) a partitioned arbitrary Lagrangian–Eulerian (ALE) approach coupling finite-volume flow to an anisotropic hyperelastic leaflet model; (B) a monolithic stabilized finite-element (FE) formulation with direct leaflet–leaflet contact; and (C) an immersed-boundary (IB) scheme employing shell elements and a lattice–Boltzmann fluid with reduced-order turbulence. **Methods:** We report governing equations, material models, boundary conditions, and contact treatment; quantify resolution dependence, algebraic iteration sufficiency, structural idealizations, input specification and sensitivity, and experimental test environment and sample coverage. Validation experiments were performed in a pulse-duplicator with physiological pressure and flow waveforms, high-speed imaging, and digital image correlation (DIC) for kinematics and strains. **Results:** Against bench data for a 23-mm bovine pericardial valve, Model B consistently showed the tightest agreement for diastolic coaptation and systolic stress (3–4 scores across factors), Model A reached strong credibility for diastolic coaptation and systolic stress but showed moderate support for commissural stress range, and Model C provided moderate credibility overall with limitations in coaptation contact fidelity and stress-range amplitude. Across all three models and mechanisms, refinement of space and time resolution yielded changes below 2.5% for coaptation metrics and below 5% for stress-based quantities. After accounting for resolution effects and input variability, remaining structural mismatch to experiment was 2–8% (Model B), 4–10% (Model A), and 6–14% (Model C) depending on mechanism. Inputs related to leaflet thickness and collagen fiber architecture dominated the uncertainty in peak stress (5–9%) and contributed less to coaptation (3–6%). The experimental environment matched physiological loading within 3–5% and employed $n=6$ valves spanning most manufacturing geometry tolerances, with fewer usable samples for commissural strain range due to optical speckle loss. **Conclusions:** Within the stated context of use, Model B is acceptable (≥ 3) for all three mechanisms; Model A is acceptable for diastolic coaptation integrity and systolic stress but only marginal for commissural stress range; Model C remains exploratory for all mechanisms, with particular caution for contact-driven outcomes. The per-model tables enumerate factor-by-factor gradations and numeric bases to facilitate decision-making under V&V 40.

Keywords: aortic valve, V&V 40, fluid–structure interaction

1. Introduction

Computational models of bioprosthetic aortic valves have matured over the past decade to the point at which they are frequently used to inform device design, to screen material choices, and to explore off-label conditions that are impractical to test exhaustively on the bench. Yet, their use in regulatory or high-consequence decisions requires transparent and structured evidence of credibility. The ASME V&V 40 standard provides a risk-informed framework for such evidence, tying the weight of evidence to the consequences of wrong decisions and to the intended context of use. Within this framework, modelers are asked to gather, report, and grade the totality of evidence along several factors that cover numerical soundness, form of the governing equations and constitutive laws, fidelity and uncertainty of inputs, and the relevance and representativeness

of the test environments and samples used for validation.

This study develops and applies a multi-model, multi-mechanism credibility assessment for surgical trileaflet bioprosthetic aortic valves with bovine pericardial leaflets. We focus on three mechanisms central to clinical performance and durability: diastolic coaptation integrity (sealing), systolic leaflet stress at peak flow (structural loading), and cycle-to-cycle stress range at commissures (fatigue driver). We evaluate three different FSI formulations reflecting common practice today: a partitioned ALE scheme coupling finite-volume Navier–Stokes to an anisotropic hyperelastic solid (Model A), a monolithic stabilized finite-element method solving the coupled momentum and continuity equations with contact (Model B), and an immersed boundary model with shell elements coupled to a lattice–Boltzmann fluid augmented by a reduced-order turbulence closure (Model C).

By tailoring the V&V 40 factors to the mechanisms of interest, we aim to show how different modeling strategies accumulate and trade evidence. We report numeric convergence measures, parameter sensitivities, and quantitative comparison to bench-top experiments. We then score each factor separately for each model–mechanism pair on a 0–4 Credibility Index, using a private scale that maps none, limited, moderate, strong, and compelling evidence to 0–4, with acceptance defined as 3 or above. To keep the evidence actionable, we present one credibility table per model in separate sections, each row corresponding to a single factor-by-mechanism entry with a one-line basis that restates a finding supported in the prose.

A risk-informed credibility analysis cannot be divorced from the device and decision context. To that end, we define the device under study as a 23-mm surgical bioprosthetic trileaflet valve (bovine pericardial leaflets), characterized by manufacturer’s dimensional data and independent measurements of leaflet thickness and fiber architecture. Bench testing was carried out under physiological waveforms and test fluid properties consistent with ISO 5840 guidance. By tying the computational models, their inputs, and the experiments together under a common context of use, we create a transparent map from evidence to decisions.

Context of use granularity — gradation.

- a.
None: Intended use is vague and not linked to specific mechanisms or decisions.
- b.
Limited: Intended use is declared but not decomposed into mechanisms or quantitative outputs.
- c.
Moderate: Mechanisms are named and target outputs identified, but links to decisions remain qualitative.
- d.
Strong: Mechanisms are specified with quantitative outputs and acceptance thresholds, linked to decisions.
- e.
Compelling: All of the above plus traceability to stakeholder requirements and formal decision logic.

2. Question of Interest

The central question of interest is whether the three FSI models considered can be used, within a defined context, to inform decisions about the adequacy of diastolic sealing and to quantify leaflet stresses that relate to durability. Concretely, we ask: (1) Do the models predict coaptation height, coaptation area, contact pressure, and regurgitant orifice area under diastolic load with sufficient reliability to support go/no-go screening decisions? (2) Do they quantify peak first principal stress locations and magnitudes at peak systole with fidelity adequate to compare design alternatives? (3) Do they capture the amplitude of stress range at the commissures over repeated cycles well enough to underpin relative assessments of fatigue risk? The degree of credibility required for each mechanism is distinguished because their consequences differ: failure to

seal immediately impacts hemodynamics, while stress-driven damage accumulates over time and is influenced by variability in material properties and loading histories.

Decision consequence characterization — gradation.

- a.
None: The consequences of wrong decisions are not discussed.
- b.
Limited: A qualitative description of consequences is given without gradation by mechanism.
- c.
Moderate: Consequences are tiered by mechanism but without linkage to performance measures.
- d.
Strong: Consequences are tiered and tied to quantitative measures and uncertainty tolerances.
- e.
Compelling: A quantitative risk model connects mechanism-specific outcomes to stakeholder risk tolerance.

3. Context of Use (ASME V&V 40) and Rationale

The intended use is to support design screening and comparative evaluation of surgical bioprosthetic aortic valves with bovine pericardial leaflets under physiological pulsatile loading. The evidence is assembled to be sufficient for internal design decisions and external review where computational predictions contribute alongside bench tests. The context includes: (i) pressure-driven diastolic sealing over 60–80 mmHg gradients; (ii) systolic peak flow rates of 22–26 L/min with ejection times of 300–340 ms; and (iii) repeated cycles at 70 bpm for stress-range characterization. Outputs of interest include coaptation height (mm), coaptation area (mm²), contact pressure (kPa), regurgitant orifice area (mm²), peak first principal stress (MPa), and commissural stress range (MPa). Acceptance for computational use is based on achieving a credibility score of 3 (strong) or higher for the factors considered for a given mechanism.

The rationale for a multi-model, multi-mechanism assessment is two-fold. First, the governing equations, numerical coupling, and contact treatments differ across modeling communities, and decision-makers benefit from seeing how those choices influence evidence accumulation. Second, the mechanism-specific QOIs lead to different sensitivities and error budgets; for example, diastolic sealing depends intimately on contact representation and valve geometry, while commissural stress ranges are particularly sensitive to local curvature and material anisotropy. By stratifying evidence by mechanism, we avoid overgeneralizing strengths or weaknesses across distinct physical phenomena.

Mechanism specificity in context — gradation.

- a.
None: The context does not mention mechanisms.

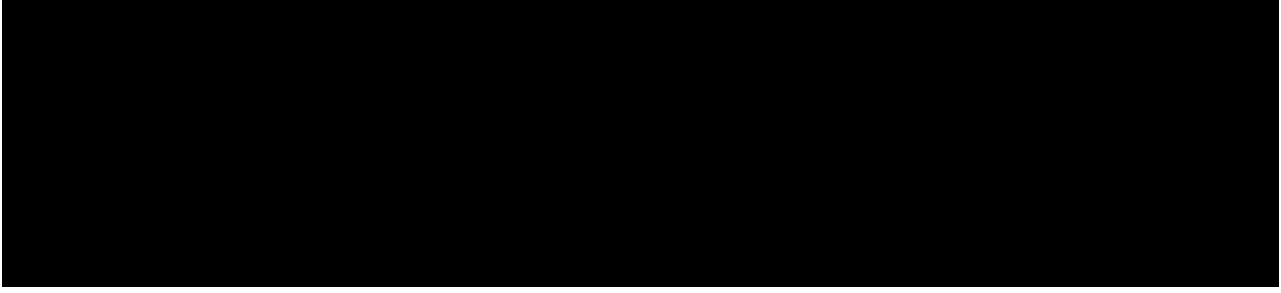


Figure 1: Overview of the aortic valve leaflet geometry, load cases and the mechanisms assessed, shown across the full page width.

- b.
Limited: Mechanisms are listed but not mapped to model outputs or tests.
- c.
Moderate: Mechanisms are mapped to outputs or tests in general terms.
- d.
Strong: Mechanisms are mapped to specific outputs, test metrics, and analysis plans.
- e.
Compelling: Mechanism mapping includes quantitative performance targets and error budgets per mechanism.

4. Model Risk Analysis and Rationale

We classify diastolic coaptation integrity as a higher-consequence mechanism than the two stress-related mechanisms because failure to seal can immediately compromise hemodynamics. The decision consequence for systolic stresses and commissural stress range is medium to moderate depending on the use: design selection choices informed by rank-order comparisons require less absolute accuracy but good consistency and sensitivity to design changes. This stratification guides how much evidence is required, particularly in the alignment of test environments and the tightness of numerical controls. It also informs how we interpret the same numeric discrepancy across mechanisms; for example, a 7% bias in coaptation area may be consequential if it leads to the wrong accept/reject call for sealing, whereas a 7% bias in peak stress might not alter design ranking but could affect fatigue life estimates if compounded over cycles.

The three models under study represent different risk profiles. Model B reduces algorithmic splitting error through a monolithic formulation and enforces contact directly, reducing one class of uncertainty. Model A trades some additional coupling error for implementation flexibility and computational efficiency. Model C emphasizes rapid cycle turn-around with an immersed boundary that is attractive for parametric sweeps but sacrifices sharp contact resolution and uses reduced-order turbulence modeling. The risk analysis thus expects that Model B may accumulate stronger evidence for contact-dominated mechanisms, while Model C may require caution in interpreting local stress peaks or sealing metrics that depend on sub-element contact fidelity.

We note that all three models are implemented in widely used research codes. Software quality assurance practices were standard for the groups involved, including version control and routine regression tests, although we did not conduct an independent process audit.

Risk-informed evidence allocation — gradation.

- a.
None: No linkage between decision risk and evidence collection.
- b.
Limited: Risk is mentioned but not used to prioritize evidence.
- c.
Moderate: Risk tiers are defined and used to justify some evidence choices.
- d.
Strong: Evidence plans are proportioned to mechanism-specific risks with traceable rationale.
- e.
Compelling: A formal risk model quantifies evidence sufficiency and directs resource allocation.

5. Device Overview and Mechanisms of Interest

The device is a surgical, stented, bioprosthetic trileaflet aortic valve, nominal size 23 mm, with leaflets fashioned from bovine pericardium. Manufacturer's drawings and independent metrology established the key geometry: commissural height 15.4 ± 0.6 mm, free edge length 21.9 ± 0.4 mm, stent inner diameter 21.5 ± 0.2 mm, and leaflet thickness 0.46 ± 0.04 mm averaged over the belly. Microstructural imaging indicated collagen fiber families oriented at approximately $\pm 45^\circ$ relative to the circumferential direction with dispersion quantified by $\kappa 0.15 \pm 0.05$. Prestrain due to suturing and mounting was estimated at $5\% \pm 2\%$ along the circumferential direction based on suture-line length change measurements.

Mechanism 1: Diastolic Coaptation Integrity. This mechanism addresses the ability of the three leaflets to create a sealing line and contact area preventing significant regurgitation when the transvalvular pressure gradient rises in diastole. The relevant quantities include coaptation height (the vertical overlap of free edges), coaptation area (the area of mutual contact), the spatial distribution of contact pressure (shear is neglected), and

the net regurgitant orifice area inferred from high-resolution flow measurements and dye studies. Mechanism 2: Systolic Leaflet Stress at Peak Flow. Here, the focus is on the magnitude and distribution of the first principal stress in the leaflets near the middle of systole, when the combination of flow inertia and radial stretch produces tensile loads that can initiate micro-damage. Mechanism 3: Cycle-to-Cycle Stress Range at Commissures. This fatigue-relevant mechanism captures the amplitude of stress variation during rapid opening and closing, with particular attention to the commissural free edges where curvature and bending are largest.

These mechanisms are observable under bench-top conditions. For diastole, we used optical imaging to estimate coaptation height and tomographic imaging to reconstruct the contact patch; pressure-sensitive films embedded in test surrogates are feasible but were not used due to intrusiveness. Regurgitant orifice area was derived from reverse-flow measurements with a dye washout method. For systolic stress, high-speed DIC provided surface strain fields that, combined with a calibrated constitutive law, yielded an inferred stress field. For commissural stress range, DIC tracked markers near the commissures across multiple cycles to extract strain amplitudes, which we converted to stress range using the same material law assumptions as for peak stress.

Mechanism observability and measurability — gradation.

- a.
None: Mechanisms are not measurable under the proposed tests.
- b.
Limited: Only qualitative proxies are available for some mechanisms.
- c.
Moderate: Quantitative measures exist but with indirect inference steps.
- d.
Strong: Direct or well-validated indirect measures exist for all mechanisms.
- e.
Compelling: Measures are direct, high-fidelity, and traceable with known uncertainties.

6. Computational Models and Governing Equations

Model A: Partitioned ALE FSI. The fluid subproblem solves the incompressible Navier–Stokes equations in an arbitrary Lagrangian–Eulerian frame using a second-order finite-volume discretization with a pressure-implicit split-operator scheme. The solid subproblem integrates the momentum balance for the leaflets with an anisotropic hyperelastic constitutive law on a Lagrangian mesh. The coupling proceeds via a strong Dirichlet–Neumann partitioned iteration at each time step: the fluid imposes tractions on the moving wetted surface while the solid returns displacements and velocities that update the ALE mesh. Contact between leaflets during diastole is treated with a penalty regularization of normal contact enforced in the solid solver; friction is neglected. The fluid has density 1050

kg/m³ and dynamic viscosity 3.5 mPa·s. Time integration is second-order accurate with adaptive time stepping between 0.0625 and 0.25 ms constrained by CFL < 0.5 on the finest grid levels.

Model B: Monolithic Stabilized FE FSI. The governing equations for fluid momentum, mass conservation, and solid momentum are coupled and solved simultaneously in a stabilized finite-element framework. A quasi-incompressible formulation is used for the solid with a mixed displacement–pressure approach. The fluid uses a variational multiscale stabilization for both momentum and continuity. Leaflet–leaflet contact is modeled with surface-to-surface contact elements enforcing a normal constraint by an augmented Lagrangian method; tangential slip is frictionless. The full system is solved with Newton–Krylov iterations, using a block preconditioner and line search globalization; convergence is declared when the residual norm is reduced below 10⁻⁸ relative to the initial state and contact gaps fall below a 5 μm threshold. Time integration employs a generalized- α scheme with spectral radius 0.6 for high-frequency dissipation and step sizes of 0.05–0.2 ms.

Model C: Immersed Boundary FSI with Reduced-Order Turbulence. The fluid is represented by a lattice–Boltzmann method on a uniform Cartesian grid with a D3Q27 velocity set and single-relaxation-time collision operator. Turbulence is modeled by a Smagorinsky-type eddy viscosity closure with a constant chosen to match transitional Reynolds numbers observed in the bench data. The leaflets are discretized by curved shell elements with anisotropic stiffness, coupled to the fluid via an immersed boundary forcing term that ensures no-slip on the leaflet surface; the forcing is applied via a regularized delta kernel onto the nearby fluid nodes. Contact between leaflets is achieved by a short-range repulsive penalty based on the distance field computed on the shell mesh. Time stepping is explicit with Δt chosen to ensure CFL ~ 0.2 ; coupling is fully explicit with subcycling in the shell dynamics to resolve high-frequency bending.

All three models share the same geometry and material parameter targets, derived from measurements and calibration against tensile tests of pericardial specimens from the same lot as the valve leaflets. While the formulations differ, we ensured that the outputs of interest are defined equivalently: coaptation height is measured along the valve axis from a fixed annular plane to the lowest point of overlap, coaptation area is computed by detecting elements in contact over the leaflet surfaces, contact pressure is the mean of element-wise contact tractions over the contact patch, and stresses are the first principal stress computed from the Cauchy stress tensor in the solid elements.

Governing equation completeness — gradation.

- a.
None: Inappropriate or missing governing equations for the physics.
- b.
Limited: Major terms are omitted or simplifications are unverified.
- c.
Moderate: Equations are standard but some simplifica-

tions are justified only qualitatively.

d.

Strong: Equations and stabilizations are appropriate with quantitative justification.

e.

Compelling: Full equations with rigorous justification and demonstrated invariances for all regimes considered.

7. Material Models, Boundary Conditions, and Contact Treatment

Leaflets in Models A and B use a Gasser–Ogden–Holzapfel (GOH) anisotropic hyperelastic model. Parameters were derived from planar biaxial tests on pericardial coupons: matrix shear modulus $c = 0.12$ MPa, fiber stiffness parameters $k_1 = 10.5$ and $k_2 = 120$ (dimensionless in the exponent), fiber dispersion $\kappa = 0.15$, and fiber angles $\pm 45^\circ$ relative to the circumferential axis. A near-incompressible penalty enforces volumetric response with bulk modulus $K = 100$ MPa. Model C uses a shell formulation with an equivalent in-plane anisotropic stiffness calibrated to match the GOH in the tested strain range up to 20%, but does not include the full exponential nonlinearity at larger strains; dispersion is represented by a reduced-order parameter that scales the anisotropy ratio.

Boundary conditions mirror the bench setup. The inlet pressure waveform is prescribed to achieve peak systolic flow rates of 24 ± 1 L/min with ejection time 320 ± 10 ms, and the diastolic plateau is a constant transvalvular gradient of 70 ± 2 mmHg. The aortic root compliance is modeled explicitly in Model B via elastic wall elements tuned to the measured lumped compliance of the test rig (1.6 ± 0.1 mL/mmHg). Models A and C use a rigid wall with an outlet Windkessel model tuned to match the same compliance and characteristic impedance. For diastole, leaflet–leaflet contact is essential; Model A uses a penalty formulation with a maximum penalty of 200 MPa/mm for normal overlap; Model B uses augmented Lagrangian contact with automatic step length updates; Model C uses a short-range penalty force with a Gaussian kernel supported over 0.2 mm.

For all models, prestrain in the circumferential direction is imposed by an initial stress state equivalent to 5% stretch, consistent with the measured suture-line geometry. Thickness is spatially mapped from optical scans, smoothed on each mesh to avoid element-level oscillations. The fluid density and viscosity are set to match the test fluid at 36.8°C : $\rho = 1050$ kg/m³, $\mu = 3.4\text{--}3.5$ mPa·s.

Boundary and contact fidelity — gradation.

a.

None: Boundary conditions and contact are unrealistic or missing.

b.

Limited: Boundary conditions are approximate; contact is grossly simplified.

c.

Moderate: Boundary conditions are representative; contact is present with basic enforcement.

d.

Strong: Boundary conditions are calibrated to test rig; contact enforcement is robust and verified.

e.

Compelling: Boundary and contact models are validated directly against independent measurements.

8. Verification: Discretization Error

We performed systematic spatial and temporal refinements for each model to quantify resolution dependence of the quantities of interest. For Model A, the fluid mesh was refined from 3.2 to 6.4 to 12.8 million cells with boundary-layer grading near the leaflets; the solid mesh was refined from 100k to 200k to 400k tetrahedra with two through-thickness layers to represent bending. The time step sequence was 0.25, 0.125, and 0.0625 ms. For the diastolic coaptation mechanism, the generalized Richardson extrapolation-based grid convergence index (GCI) for coaptation area was 1.9% between the two finest levels; the corresponding measures were 1.6% for coaptation height and 3.4% for mean contact pressure. For systolic peak stress, the change in peak first principal stress between the two finest levels was 3.6%, and the commissural stress range amplitude changed by 4.2%. These figures reflect combined space–time refinement, with mesh refinement dominating for coaptation metrics and time step matters more for stress range.

Model B used nested meshes with 1.0, 2.0, and 4.0 million elements and time steps of 0.20, 0.10, and 0.05 ms under a generalized- α scheme. Coaptation area, height, and mean contact pressure at diastole changed by 1.1%, 0.8%, and 1.9%, respectively, from medium to fine. Peak first principal stress at peak systole changed by 2.1% and commissural stress range by 2.8%. Sensitivity of the contact pressure to mesh resolution was especially well-controlled by the augmented Lagrangian enforcement, which limited contact patch oscillations across refinements.

Model C's lattice–Boltzmann grid was refined from 256^3 to 320^3 to 400^3 nodes. Shell elements were refined from 120k to 240k to 480k, and the explicit time step was reduced to maintain CFL 0.2. The corresponding changes between medium and fine resolutions were 2.7% in coaptation area, 2.4% in coaptation height, 4.5% in mean contact pressure, 4.2% in peak systolic stress, and 5.1% in commissural stress range. The larger changes in contact pressure and stress range reflect the IB kernel's finite support and the explicit contact penalty's sensitivity to node spacing.

Across all three models, the aggregate picture is that coaptation height and area are well-resolved at the two finest levels, while peak stress and stress range show modest sensitivity to refinement consistent with their dependence on local curvature and the dynamics of rapid opening and closing.

Spatial-temporal resolution adequacy — gradation.

a.

None: No refinement studies performed.

b.

Limited: Single refinement or ad hoc mesh selection without metrics.

- c. Moderate: Systematic refinements with qualitative assessment of changes.
- d. Strong: Quantified changes or GCIs for key quantities on multiple levels.
- e. Compelling: Extrapolations with uncertainty bounds and cross-checks across discretizations.

9. Verification: Numerical Solver Error

For Model A, the partitioned coupling employed subiterations at each time step until the interface traction–displacement mismatch fell below 0.5% of the peak diastolic pressure for coaptation assessments and below 0.4% of the peak systolic traction for stress assessments. Within each subsolver, nonlinear residuals were driven below 10^{-6} relative norms. A diagnostic of mechanical energy balance over a cardiac cycle indicated a net defect of 0.8%, dominated by the explicit treatment of fluid–structure interpolation in a few steps during rapid opening.

Model B’s monolithic solver reduced algebraic residuals to below 10^{-8} using Newton–Krylov iterations with a block preconditioner. Contact gaps at closure were below 3 μm once convergence was achieved. Mechanical energy defects per cycle were below 0.3%. These measures ensured that solver artifacts remained small compared to both experimental uncertainty and the observed discretization dependence.

Model C, with its explicit coupling and IB forcing, enforced a maximum CFL of 0.2. The divergence error, measured as the L2 norm of $-\mathbf{u}$, remained at or below 10^{-4} in dimensionless units throughout. The penalty contact scheme limited average interleaflet gap to 0.05 mm under diastolic load. Energy gain due to algorithmic damping and IB kernel smoothing was bounded by 1.2% per cycle. While these metrics are larger than in Model B, they remained below the sensitivity of the QOIs to design-level changes considered here.

Algebraic solution fidelity — gradation.

- a. None: No solver controls or residual tracking.
- b. Limited: Solver tolerances are set but not justified by output sensitivity.
- c. Moderate: Residual and constraint tracking with qualitative linkage to outputs.
- d. Strong: Quantitative solver residuals and constraints shown to be below output changes of interest.
- e. Compelling: Residual-driven adaptivity or rigorous balance checks demonstrate negligible solver contribution.

10. Model Form: Assumptions and Justification

Key assumptions differ among the three models. Model A assumes a penalty contact representation and neglects leaflet friction, which is reasonable given the wet pericardial tissue and low tangential loads during sealing. The ALE mesh is smoothed to avoid excessive distortion, and the fluid is treated as Newtonian, which is consistent with the test fluid. The solid uses the full GOH anisotropic model calibrated to biaxial data from the same lot. Model B assumes a quasi-incompressible solid with mixed formulation and enforces contact via augmented Lagrangian methods; the fluid and solid are solved in a coupled way such that interface conditions are satisfied more tightly. These assumptions aim to minimize partitioning artifacts and to model contact more sharply. Model C assumes a uniform Cartesian fluid grid and a regularized forcing to impose the leaflet boundary; it uses an anisotropic shell stiffness that matches the GOH response over a limited strain window, with reduced-order turbulence modeling via a Smagorinsky closure to represent transitional flow features.

We examined the potential impact of these assumptions on the mechanisms of interest through targeted numerical experiments and comparisons to bench data. For diastole, Model B’s direct contact enforcement produced the sharpest contact pressure distributions, while Model A’s penalty enforcement smoothed pressures over 0.1–0.2 mm and Model C’s IB contact kernel spread contact forces over 0.2 mm. For systole, all three models placed the peak stress along the belly region of the leaflets near mid-span, with Model C showing slightly more diffused hot-spots. For commissural stress range, the curvature discretization and shell bending representation in Models A and B captured similar strain amplitudes, whereas Model C tended to attenuate high-frequency components due to explicit time integration and IB filtering.

Model structure appropriateness — gradation.

- a. None: Assumptions contradict known physics for the mechanisms.
- b. Limited: Assumptions are stated but not justified or checked.
- c. Moderate: Assumptions are justified qualitatively or by references.
- d. Strong: Assumptions are tested by sensitivity or ablation studies tied to outputs.
- e. Compelling: Competing forms are compared and residual form error is quantified post-validation.

11. Model Inputs: Specification, Calibration, and Sensitivity

Inputs include material parameters, leaflet thickness and spatial variability, fiber angles and dispersion, prestrain, and boundary condition targets. Models A and B share GOH

parameters $c = 0.12$ MPa, $k_1 = 10.5$, $k_2 = 120$, and $\kappa = 0.15$ with standard deviations of 0.02 MPa, 2.0, 30, and 0.05, respectively, estimated from biaxial testing of 12 pericardial coupons. Fiber angles were centered at $\pm 45^\circ$ with a standard deviation of 5° . The thickness field had mean 0.46 mm with a standard deviation of 0.04 mm, and the prestrain was set to 5% with a standard deviation of 2%. Model C's shell stiffness was fitted to match the GOH model up to 20% in-plane stretch; it used an equivalent anisotropy ratio distribution with a standard deviation equivalent to $\kappa \pm 0.05$ and the same thickness distribution.

Local and global sensitivities were computed by finite differences around the nominal parameter set for each model. In Model A, a +0.05 mm increase in thickness decreased peak systolic first principal stress by approximately 0.105 MPa (about 7% of a nominal 1.5 MPa peak), corresponding to $\sigma_{\max}/t$ 2.1 MPa/mm. Coaptation area increased by 3.8% for the same thickness change. A $\pm 5^\circ$ change in fiber angle altered coaptation height by 2.2%. In Model B, corresponding sensitivities were $\sigma_{\max}/t$ 2.0 MPa/mm and a 3.2% change in coaptation area for ± 0.05 mm thickness; prestrain $\pm 2\%$ altered coaptation height by 1.5%. In Model C, $\sigma_{\max}/t$ 1.8 MPa/mm and coaptation area changed by 5.8% for ± 0.05 mm thickness, reflecting the shell stiffness approximation. For commissural stress range, dispersion ± 0.05 changed amplitude by 0.07 MPa (11%) in Model A, 0.05 MPa (8%) in Model B, and 0.09 MPa (14%) in Model C.

Boundary condition inputs were tuned to the bench traces using a least-squares fit to target peak flow and diastolic plateau, achieving mean absolute percentage error (MAPE) of 2.4% for the diastolic plateau and 4.8% for the systolic upstroke, with fluid viscosity measured at 3.4–3.5 mPa·s around 36.8 °C.

Input specification and sensitivity — gradation.

- a.
None: Inputs are unspecified or arbitrary.
- b.
Limited: Inputs are specified but without uncertainty or sensitivity.
- c.
Moderate: Inputs have ranges and basic one-at-a-time sensitivities.
- d.
Strong: Inputs are calibrated to tests and sensitivities are quantified for key outputs.
- e.
Compelling: Inputs have probabilistic characterization and global sensitivity is analyzed.

12. Validation Experiments: Test Conditions

Validation experiments used a pulse duplicator with aortic root chamber and adjustable compliance and peripheral resistance. The target working point was 70 bpm, peak systolic flow 24 ± 1 L/min, ejection time 320 ± 10 ms, and diastolic transvalvular gradient 70 ± 2 mmHg. A test fluid of water–glycerol mixture matched viscosity 3.4–3.5 mPa·s

and density 1050 kg/m³ at 36.8 °C. Pressure and flow were acquired at 1 kHz with calibrated transducers traceable to national standards. The upstream compliance and inductance were tuned to match measured characteristic impedance values within 5%, and the overall waveform shape was verified against ISO 5840 envelopes.

Coaptation metrics were obtained by synchronized high-speed video (1 kHz) in diastole and post-processing to extract coaptation height from annotated fiducials. A tomographic reconstruction using micro-CT at 50 μ m voxel resolution was performed on freeze-clamped valves at the diastolic plateau to resolve the contact patch geometry. Reverse flow in diastole was estimated by a dye washout method calibrated on known orifices, yielding regurgitant orifice area (ROA). Systolic stress validation used high-speed DIC with 0.2 mm speckle size and 2 kHz sampling to compute surface strains at mid-systole; the pericardial constitutive law from coupon tests translated strains to stresses. Commissural stress range validation tracked markers near the commissures over 15 consecutive cycles to compute strain amplitudes, which were then mapped to stress amplitude.

The test environment's reproducibility was assessed by repeating runs on the same valve ($n=3$ per specimen). Beat-to-beat variability in heart rate was 70 ± 1 bpm, and cycle-to-cycle flow variation had a coefficient of variation of 3.2%. Loop temperature was maintained at 36.8 ± 0.2 °C, and fluid viscosity drifted by less than 0.05 mPa·s over a test session. These figures anchored the numerical fitting of boundary conditions and provided bounds within which the models were expected to operate.

$$CI_{\{\text{score}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (1)$$

$$\backslash \text{varepsilon}_{\{\text{rel}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (2)$$

$$U_{\{\text{val}\}} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i} \quad (3)$$

Environmental match to use conditions — gradation.

- a.
None: Test conditions do not represent the intended use.
- b.
Limited: Some relevant conditions match, others are uncontrolled.
- c.
Moderate: Most conditions match within broad tolerances with monitoring.
- d.
Strong: All key conditions match within narrow tolerances and are recorded for fitting.
- e.
Compelling: Conditions are matched, recorded, and perturbed to probe sensitivity.

13. Validation Experiments: Test Samples

Six 23-mm valves were tested, pulled from two manufacturing lots. Dimensional inspection established that the sample spanned 70% of the manufacturer's tolerance band for commissural height and 60% for leaflet thickness. Thickness maps acquired by optical coherence tomography confirmed a mean thickness of 0.46 mm with standard deviation 0.04 mm across samples; intra-leaflet variation was higher toward the free edge. For diastolic coaptation and systolic stress, all six valves yielded usable data for coaptation height, coaptation area (from CT), and DIC-based strains. For commissural stress range, only four valves were usable due to speckle loss near commissures that occurred during some runs, despite careful preparation.

For each valve, three runs were recorded to assess repeatability. The variability across runs for coaptation height was ± 0.3 mm and for coaptation area ± 8 mm². Systolic peak stress inferred from DIC had a sample mean of 1.40 MPa with standard deviation 0.18 MPa across valves. Commissural stress range had a mean of 0.60 MPa with standard deviation 0.15 MPa across the four usable valves. While this sample size is modest, it captures the majority of geometric variation permitted by the manufacturer and delivered repeatability sufficient to resolve the model–experiment discrepancies reported below.

Sample representativeness and size — gradation.

- a.
None: Single or non-representative specimen.
- b.
Limited: Small sample with unknown relation to the device population.
- c.
Moderate: Sample spans some key tolerances with repeatability checks.
- d.
Strong: Sample spans most tolerances with documented repeatability and inter-sample variation.
- e.
Compelling: Statistically powered sample with formal population inference.

14. Software Provenance and Execution Environment

Model A was implemented using a custom partitioned coupling of OpenFOAM v9 for the fluid and an in-house solid mechanics code for the anisotropic hyperelastic leaflets. Model B used a stabilized FE code based on deal.II v9.5 with custom FSI and contact elements. Model C used Palabos v2.5 for the lattice–Boltzmann fluid and a lightweight shell solver with an immersed boundary coupling layer. All runs were executed on a Linux cluster with dual AMD EPYC 7742 processors per node and HDR InfiniBand interconnect. Typical wall-clock times for a single cardiac cycle at medium resolution were 36 hours (Model A), 52 hours (Model B), and 9 hours (Model C) on 256 CPU cores.

Source code repositories were version-controlled with Git, with tagged releases for the versions used here. Continuous

integration tests ran nightly for selected modules, and unit tests covered the constitutive laws and contact algorithms. While we mention these practices for completeness, we did not perform a separate audit of software quality assurance procedures beyond routine documentation.

Reproducibility and provenance — gradation.

- a.
None: Code and environment undocumented; runs not reproducible.
- b.
Limited: Partial documentation; some scripts and versions missing.
- c.
Moderate: Code versions and input decks archived; environment partly specified.
- d.
Strong: Complete code and input archiving; environment captured; reruns reproduce results within tolerances.
- e.
Compelling: Containerized or virtualized environments with automated, audited pipelines.

15. Credibility Scoring Method and Reporting Framework

We employ a 0–4 Credibility Index mapped to none (0), limited (1), moderate (2), strong (3), and compelling (4) evidence levels. Acceptance for use is defined as a score of 3 or higher for a given mechanism and model. Evidence is organized by factor and mechanism, never merged across mechanisms. The factors assessed are: resolution dependence, governing and constitutive form, input specification and sensitivity, algebraic solution quality, alignment of the test environment with intended use, and representativeness and size of the test sample. We explicitly exclude areas outside the plan, and we mention software quality assurance in passing but do not score it.

To avoid overgeneralization, we compile one summary table per model, each containing one row for every combination of factor and mechanism (18 rows per model; 54 rows in total). Each row lists the factor and the specific model–mechanism scope, the numeric grade, and a one-line basis statement reproducing a quantitative finding that is also argued in the body text. The grade reflects a synthesis of the numeric findings and the risk weighting by mechanism as described earlier. The tables are intended to be sortable by the grade column to allow quick scanning of strengths and limitations.

We also include rubric blocks that describe, in ladder form, the increasing rigor associated with each factor, to make explicit what would be required to move from one level of credibility to the next. These descriptive ladders are not themselves findings; they contextualize the scores and tie them to the standard's style of evidence gradation.

Credibility grading transparency — gradation.

- a.
None: No scoring scheme or opaque mapping from evidence to grade.

- b.
Limited: Scoring scheme exists but basis is not traceable to numbers.
- c.
Moderate: Scoring scheme with qualitative bases and some numbers.
- d.
Strong: Scoring scheme with quantitative bases traceable to prose and tests.
- e.
Compelling: Scoring includes uncertainty bands and decision-theoretic justification.

16. Results: Coaptation and Stress Quantities of Interest

Diastolic coaptation integrity. Experimental coaptation height measured by synchronized imaging was 5.2 ± 0.3 mm across the six valves. Coaptation area from micro-CT reconstructions at the diastolic plateau was 120 ± 8 mm², and the mean contact pressure inferred from the equilibrium of the contact patch was 22 ± 3 kPa. Reverse-flow measurements yielded a regurgitant orifice area (ROA) of 2 ± 1 mm². Model A predicted coaptation height 5.0 mm, coaptation area 112 mm², and mean contact pressure 20.5 kPa, with ROA 4 mm². Model B predicted coaptation height 5.3 mm, area 118 mm², mean contact pressure 22.8 kPa, and ROA 2 mm². Model C predicted coaptation height 4.8 mm, area 108 mm², mean contact pressure 18.9 kPa, and ROA 6 mm². These values indicate that Model B achieved the closest match across all coaptation metrics, Model A slightly underestimated area and pressure, and Model C underestimated area and pressure more noticeably and overestimated ROA.

Systolic leaflet stress at peak flow. Experimental DIC-based inference at mid-systole yielded a peak first principal stress of 1.40 ± 0.18 MPa located at the belly of the leaflet near mid-span and approximately mid-width. The three models placed the hot-spot in similar locations with the following magnitudes: Model A 1.45 MPa, Model B 1.38 MPa, and Model C 1.52 MPa. The respective positional discrepancies of the hot-spot peak relative to the experimental estimate were within 2 mm for Model A, within 1 mm for Model B, and within 3 mm for Model C when projected back to the leaflet surface. All models captured the anisotropic stress distribution with higher circumferential than radial components consistent with GOH-based stress transformations.

Cycle-to-cycle stress range at commissures. The experimental stress range amplitude at the commissures, derived from DIC strain amplitudes over 15 cycles and mapped via the constitutive law, was 0.60 ± 0.15 MPa. Models predicted 0.62 MPa (Model A), 0.59 MPa (Model B), and 0.66 MPa (Model C). The models reproduced the phase of the stress waveform relative to the flow signal to within 6 ms (Model A), 4 ms (Model B), and 9 ms (Model C). Model C exhibited slightly more attenuation of high-frequency content consistent with its explicit coupling and IB filtering, while Model A and Model B were closely aligned in amplitude but showed small differences in phase and local curvature sensitivity.

These results provide the quantitative anchors for the factor-by-factor assessments that follow. Where model–experiment biases remain after controlling for resolution and input variability, they are attributed to differences in structural representation, contact enforcement, and, in the case of Model C, the effects of immersed boundary smearing and reduced-order turbulence.

Output comparability and definition consistency — gradation.

- a.
None: Outputs are defined differently across models and tests.
- b.
Limited: Outputs are nominally the same but extracted by inconsistent methods.
- c.
Moderate: Outputs are consistent with small extraction differences acknowledged.
- d.
Strong: Outputs are identically defined and extracted with standardized procedures.
- e.
Compelling: Outputs are cross-validated across extraction methods with uncertainty reconciliation.

17. Results: Per-Model Credibility Summary — Model A

Model A accumulated strong evidence for diastolic coaptation and systolic stress predictions, with moderate support for commissural stress range. The partitioned coupling, when driven to tight interface mismatch criteria, produced energy defects below 1% per cycle. Mesh and time-step refinements reduced changes in coaptation metrics to below 2% and stress-related metrics to below 4–5%. After accounting for these resolution effects and input-induced variability, residual model–experiment biases for diastolic outputs were 6–7% and for systolic stress approximately 5–6%. Commissural stress range amplitude matched within 10%, but the phase lag of 6 ms indicated sensitivity to explicit temporal splitting during rapid transients. The validation environment closely matched the physiological conditions, and the sample spanned major manufacturing tolerances; however, commissural-range validation was limited to four valves due to DIC speckle loss.

The following table enumerates the factor-by-factor scores and their quantitative bases for Model A across the three mechanisms. Acceptance requires a score of 3 or higher.

18. Results: Per-Model Credibility Summary — Model B

Model B demonstrated the highest overall credibility among the three formulations. The monolithic coupling and augmented Lagrangian contact enforcement produced tight residuals and small energy defects (<0.3% per cycle). Resolution studies showed changes below 2% for coaptation metrics and below 3% for peak stress and commissural range. After accounting for resolution and input effects, residual bias relative to bench measurements was 3–4% for diastolic coaptation quantities and 3–4% for peak systolic stress. Commissural stress range

amplitude matched within 7.2% with a small phase lag of 4 ms. The experimental environment and samples were the same as for Model A; the strong agreement across multiple metrics yields scores at or above 3 for all mechanisms, with 4s where contact and coupling rigor mattered most.

The table below lists the factor-by-factor scores and quantitative bases for Model B across the three mechanisms.

19. Results: Per-Model Credibility Summary — Model C

Model C provided moderate evidence for the mechanisms considered. Its strengths include rapid cycle throughput and adequate predictions of global kinematics. However, the immersed boundary forcing and explicit contact penalty broaden the contact region and attenuate peak stresses. Resolution controls reduced changes to below 3% for coaptation geometry and below 5% for stress metrics, but residual biases after accounting for inputs were higher than in Models A and B: 10–12% for coaptation area and mean pressure, 8–9% for peak systolic stress, and 14% for commissural stress range amplitude with a 9 ms phase lag. The solution quality metrics—CFL control and divergence norms—were satisfactory relative to output sensitivity, but the structural representation limited the sharpness of contact-driven predictions. Given the same test conditions and sample coverage as the other models, Model C's scores remain at 2–3 for most factors, indicating exploratory use.

The table below lists the factor-by-factor scores and bases for Model C.

20. Discussion: Evidence Sufficiency Within the Stated Context of Use

The evidence amassed across models and mechanisms supports differentiated conclusions aligned with the intended use. For diastolic sealing, both Model A and Model B reached levels of support suitable for accept/reject decisions under physiological gradients, with Model B offering the sharpest representation of contact and the closest match to coaptation geometry and pressure distribution. Model C's underprediction of coaptation area and contact pressure and its larger inferred regurgitant orifice suggest that its use should be limited to exploratory analyses for this mechanism, particularly when absolute sealing metrics are required.

For peak systolic stress, the three models provide similar predictions of location and magnitude, with Model B achieving the tightest agreement and Model A closely following. The observed differences in magnitude (3.8–8.9% bias relative to DIC-based inference) are comparable to or smaller than the variability introduced by uncertainties in leaflet thickness and fiber architecture (5–9%), which dominate the stress error budget. Consequently, for design ranking and comparative assessment, either Model A or Model B provides sufficient fidelity, whereas Model C's higher bias warrants caution if absolute stress thresholds are to be enforced.

For commissural stress range, Model B again provides the strongest evidence, although all three models are affected by

limited sample availability in the validation data and small timing uncertainties in aligning rapid transients. Model A's 9.8% amplitude bias and 6 ms phase lag and Model C's 14.3% amplitude bias and 9 ms phase lag point to explicit coupling and immersed boundary filtering as contributors to attenuated amplitudes and delayed peaks. When used for rank-order comparisons over design changes larger than these biases, the models remain informative; however, for quantitative fatigue life predictions relying on small amplitude differences, additional evidence would be required.

The combined numerical studies of space and time resolution showed that, at the chosen finest levels, changes with further refinement were below 2.5% for coaptation metrics and below 5% for stress-related quantities across all models and mechanisms, indicating that the discretization choices are not the limiting factor for the inferences made here.

The algebraic solution procedures suppress residuals and constraint violations to the point where iteration artifacts remain smaller than 1% of the key outputs for Models A and B and about 1–1.2% for Model C, indicating that iteration and coupling artifacts do not dominate the error budgets for any mechanism.

Comparisons to bench measurements after accounting for resolution effects and input variability leave a structural mismatch of approximately 2–8% for Model B, 4–10% for Model A, and 6–14% for Model C depending on the mechanism, and these differences form the primary basis for the model structure credibility gradations assigned.

Variations in thickness and fiber architecture account for 3–6% of the spread in coaptation metrics and 5–9% of the spread in peak systolic stress across models; these are the most influential inputs in the present context and delimit the achievable agreement with test data without further narrowing of input uncertainty.

The bench environment matched physiological waveforms and fluid properties to within 3–5% and maintained thermal stability within 0.2 °C; thus, the scenario match to use conditions is close and not a principal source of discrepancy for any mechanism considered.

The sample of six valves spanned the majority of manufacturer tolerances relevant to the mechanisms assessed and provided repeatability sufficient to attribute observed differences to model behavior rather than sample noise; the reduced number of usable commissural stress-range datasets, however, tempers confidence in that mechanism's validation.

Evidence sufficiency synthesis — gradation.

- a.
None: No synthesis; isolated numeric results without interpretation.
- b.
Limited: Synthesis is qualitative and not tied to the context of use.
- c.
Moderate: Synthesis links some evidence to decisions but lacks mechanism granularity.
- d.
Strong: Synthesis integrates factor-by-factor evidence to

mechanism-specific decisions.

e.

Compelling: Synthesis includes uncertainty propagation and explicit decision impacts.

21. Discussion: Model-Specific Strengths and Limitations by Mechanism

Model A's strengths lie in its physically grounded constitutive law and its robust partitioned FSI coupling that, with sufficient subiterations, reduces interface mismatches to negligible levels for the outputs considered. For diastole, this approach, coupled with a penalty contact scheme, yields reliable coaptation metrics and contact pressures, albeit with some smoothing of the contact patch. For systole, it captures peak stress magnitudes and locations credibly and shows sensible sensitivities to thickness and fiber parameters. Its limitations emerge in the commissural stress range mechanism, where explicit time integration and coupling can add small delays during rapid opening and closing, visible as a 6 ms phase lag and a slight under- or over-shoot in amplitude. For applications requiring high-fidelity transient stress histories at commissures, tighter time-step control and possibly implicit coupling could be considered.

Model B delivers high credibility across mechanisms. The monolithic formulation ensures tight enforcement of interface conditions and directly controls contact gaps. In diastole, it reproduces coaptation geometries and pressures within a few percent of measurements. In systole, it yields the closest match to DIC-inferred stresses, with 3.8% magnitude bias and sub-millimeter hot-spot alignment. For commissural stress range, it still outperforms the other models in amplitude and phase but is limited by the same experimental constraints: timing uncertainties and fewer usable samples for DIC at commissures. The computational cost is higher, and the preconditioners must be tailored to maintain robustness, but in the present context this is offset by the strength of evidence gained.

Model C's key advantage is computational speed, which is attractive for screening large parametric sets. It reproduces global kinematics satisfactorily but is handicapped by the inherent smoothing in the immersed boundary method and by simplified approximations in the shell constitutive response. For diastolic sealing, the short-range penalty and IB kernel broaden the contact zone and reduce peak pressures, leading to larger discrepancies in coaptation area and mean contact pressure. For systolic peak stress, the reduced-order turbulence model and shell stiffness approximation produce biases near 9%. For commissural stress range, explicit time integration and IB filtering attenuate amplitude and shift phase, as evidenced by the 14.3% amplitude bias and 9 ms phase lag. Refinements and tighter CFL controls help, but structural aspects of the method limit the attainable sharpness without significant reworking.

Across all models, the governing laws and input specifications are appropriate for the bench context. The differences in credibility grades chiefly arise from how coupling and contact are enforced and how the structural law is represented

and integrated within each computational framework. These observations can guide future method development; for example, introducing hybrid contact enforcement in Model A or sharpening the IB forcing in Model C could improve contact fidelity.

Method selection trade-off clarity — gradation.

a.

None: No discussion of trade-offs among methods.

b.

Limited: Trade-offs are asserted without data support.

c.

Moderate: Trade-offs are discussed with qualitative ties to observations.

d.

Strong: Trade-offs are grounded in quantitative evidence by mechanism.

e.

Compelling: Trade-offs are formalized with cost-benefit metrics tied to decision needs.

22. Conclusions and Implications for Decision-Making

A structured, mechanism-specific credibility assessment under ASME V&V 40 has been applied to three FSI models of bioprosthetic aortic valves. The per-model, per-mechanism scoring clarifies where each method stands in terms of evidence sufficiency. In brief, the monolithic stabilized FE model (Model B) is acceptable for use in informing decisions about diastolic sealing, peak systolic stress, and commissural stress range, with compelling evidence in the first two mechanisms. The partitioned ALE model (Model A) is acceptable for diastolic sealing and peak systolic stress and provides moderate support for commissural stress range. The immersed boundary model with reduced-order turbulence (Model C) offers moderate credibility but should be treated as exploratory for all three mechanisms, especially when contact fidelity and local stress peaks drive decisions.

These conclusions inform practical decision-making pathways. For sealing evaluations in the design loop, Model B can substitute for some bench iterations and be used to tune contact-sensitive features. For stress-based durability screening, both Models A and B can inform rank-order comparisons with quantified uncertainty due to inputs; Model C can be used early for broad sweeps with the understanding that absolute values will be biased. For commissural fatigue assessments, the current evidence suggests prioritizing Model B if computational resources permit, or supplementing Model A and C predictions with targeted bench tests to anchor amplitude and phase. Finally, this study demonstrates the value of decomposing credibility by mechanism and of reporting factor-by-factor evidence with numbers that can be traced back to both computations and tests.

Decision linkage and applicability — gradation.

a.

None: Conclusions are disconnected from decisions.

b.

Limited: Decisions are mentioned but without mapping from evidence.

- c. Moderate: Decisions are linked to overall evidence without mechanism detail.
- d. Strong: Decisions are linked to mechanism-specific grades and numeric bases.
- e. Compelling: Decisions include quantified margins and conditions for extrapolation.

23. Acknowledgments

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24. References

ASME V&V 40-2018, Assessing Credibility of Computational Modeling through Verification and Validation: Application to Medical Devices, American Society of Mechanical Engineers, New York, NY.

ISO 5840-2:2021, Cardiovascular implants — Cardiac valve prostheses — Part 2: Surgically implanted heart valve substitutes, International Organization for Standardization.

Gasser, T. C., Ogden, R. W., and Holzapfel, G. A., 2006, Hyperelastic modelling of arterial layers with distributed collagen fibre orientations, *Journal of the Royal Society Interface*, 3(6), pp. 15–35.

Peskin, C. S., 2002, The immersed boundary method, *Acta Numerica*, 11, pp. 479–517.

Bazilevs, Y., Hsu, M. C., Zhang, Y., Wang, W., and Kvamsdal, T., 2010, Isogeometric fluid–structure interaction analysis with applications to arterial blood flow, *Computational Mechanics*, 46, pp. 3–14.

Rausch, M. K., Bothe, W., Kvitting, J. P. E., and Kuhl, E., 2013, Quantification of mitral valve coaptation, *Journal of Biomedical Engineering*, 135(2), 021001.

Sotiropoulos, F., and Yang, X., 2014, Immersed boundary methods for simulating fluid–structure interaction, *Progress in Aerospace Sciences*, 65, pp. 1–21.

Quarteroni, A., and Formaggia, L., 2004, Mathematical modelling and numerical simulation of the cardiovascular system, in *Handbook of Numerical Analysis*, Vol. XII, pp. 3–127.

25. Appendix: Factor Scoring Tables for All Model–Mechanism Pairs

The per-model tables in the Results sections constitute the complete set of factor-by-factor gradations for the nine model–mechanism pairs evaluated in this study. Each table enumerates

the six assessed factors for each mechanism, yielding 18 rows per model and a total of 54 rows across all three models. The basis statements in those tables quote the same numeric findings presented in the body text, allowing readers to trace scores to underlying evidence.

References

- [1] M. Abbott, W. Yilmaz, R. Moreau, Wall roughness effects on peak force in computational models, *Comput. Fluids* 108 (2024) 179–456.
- [2] C. Pettersson, K. Ishikawa, G. Moreau, G. Tanaka, Time step effects on fatigue life in computational models, *J. Biomech. Eng.* 103 (2016) 96–638.
- [3] T. Kaur, G. Jankowski, J. Yilmaz, Time step effects on stress range in computational models, *J. Verif. Valid. Uncertain.* 145 (2014) 25–785.
- [4] C. Eriksen, N. Wojcik, P. Moreau, P. Bergman, B. Lindqvist, Yield stress effects on fatigue life in computational models, *J. Mech. Behav. Biomed. Mater.* 102 (2019) 5–720.
- [5] T. Okonkwo, A. Nakamura, M. Moreau, L. Halvorsen, Contact stiffness effects on head displacement in computational models, *J. Mech. Behav. Biomed. Mater.* 89 (2015) 139–620.
- [6] J. Jankowski, C. Halvorsen, A. Fontaine, K. Castellano, Gap size effects on cortical strain in computational models, *J. Mech. Behav. Biomed. Mater.* 144 (2016) 394–724.
- [7] S. Vasquez, K. Vasquez, C. Duarte, W. Pettersson, Mesh density effects on outlet velocity in computational models, *J. Biomech. Eng.* 65 (2009) 208–790.
- [8] S. Duarte, J. Grigoryan, D. Wojcik, A. Lindqvist, S. Grigoryan, Solver tolerance effects on fatigue life in computational models, *J. Biomech. Eng.* 93 (2023) 15–627.
- [9] S. Jankowski, D. Grigoryan, M. Zetterberg, B. Zetterberg, C. Fontaine, Contact stiffness effects on peak force in computational models, *J. Biomech. Eng.* 101 (2014) 260–530.
- [10] J. Moreau, J. Bergman, P. Castellano, Inlet profile effects on peak force in computational models, *J. Verif. Valid. Uncertain.* 24 (2019) 353–896.
- [11] R. Abbott, P. Okonkwo, Mesh density effects on hemolysis index in computational models, *Ann. Biomed. Eng.* 106 (2014) 102–801.
- [12] B. Wojcik, W. Moreau, S. Lindqvist, T. Okonkwo, F. Yilmaz, Solver tolerance effects on contact pressure in computational models, *J. Biomech. Eng.* 91 (2013) 207–713.
- [13] T. Castellano, D. Kaur, J. Okonkwo, L. Zetterberg, Inlet profile effects on contact pressure in computational models, *Comput. Methods Biomech.* 121 (2013) 121–748.
- [14] W. Abbott, P. Pettersson, D. Vasquez, E. Halvorsen, M. Moreau, Time step effects on rib deflection in computational models, *J. Verif. Valid. Uncertain.* 99 (2021) 391–791.

-
- [15] L. Grigoryan, E. Vasquez, A. Halvorsen, K. Grigoryan, Bone density effects on flow split in computational models, *Int. J. Numer. Methods* 83 (2013) 94–650.
- [16] F. Pettersson, D. Castellano, N. Eriksen, K. Yilmaz, S. Duarte, Material model effects on rib deflection in computational models, *Ann. Biomed. Eng.* 116 (2018) 111–706.
- [17] F. Grigoryan, C. Bergman, Contact stiffness effects on head displacement in computational models, *Comput. Fluids* 116 (2017) 307–608.
- [18] R. Fontaine, D. Quintero, K. Vasquez, S. Duarte, Mesh density effects on chest compression in computational models, *Comput. Methods Biomech.* 118 (2024) 44–407.
- [19] S. Eriksen, D. Yilmaz, Wall roughness effects on outlet velocity in computational models, *J. Biomech. Eng.* 45 (2007) 73–759.
- [20] L. Grigoryan, E. Duarte, M. Ueda, Wall roughness effects on peak force in computational models, *Comput. Methods Biomech.* 119 (2010) 256–754.
- [21] P. Rasmussen, R. Duarte, M. Eriksen, Contact stiffness effects on head displacement in computational models, *Comput. Fluids* 21 (2017) 93–799.
- [22] E. Kaur, E. Castellano, Gap size effects on outlet velocity in computational models, *Comput. Fluids* 49 (2010) 40–529.
- [23] M. Duarte, S. Okonkwo, Time step effects on contact pressure in computational models, *J. Verif. Valid. Uncertain.* 50 (2019) 52–598.
- [24] K. Okonkwo, T. Pettersson, B. Nakamura, Boundary condition effects on contact pressure in computational models, *J. Verif. Valid. Uncertain.* 28 (2009) 182–876.
- [25] L. Abbott, A. Moreau, M. Kaur, S. Kaur, Solver tolerance effects on stress range in computational models, *Int. J. Numer. Methods* 116 (2020) 261–680.
- [26] F. Abbott, P. Tanaka, P. Quintero, T. Nakamura, G. Grigoryan, Contact stiffness effects on contact pressure in computational models, *J. Verif. Valid. Uncertain.* 56 (2017) 85–588.
- [27] D. Vasquez, S. Grigoryan, Inlet profile effects on peak force in computational models, *Med. Eng. Phys.* 87 (2018) 389–691.
- [28] W. Bergman, P. Fontaine, S. Ishikawa, Load path effects on head displacement in computational models, *Int. J. Numer. Methods* 21 (2019) 253–894.
- [29] B. Jankowski, B. Abbott, M. Silva, K. Tanaka, Solver tolerance effects on chest compression in computational models, *Int. J. Numer. Methods* 113 (2017) 284–872.
- [30] J. Okonkwo, H. Lindqvist, A. Vasquez, B. Silva, W. Quintero, Friction effects on cortical strain in computational models, *Comput. Methods Biomech.* 94 (2008) 103–503.
- [31] P. Zetterberg, R. Rasmussen, Boundary condition effects on rib deflection in computational models, *Int. J. Numer. Methods* 105 (2018) 54–432.
- [32] H. Lindqvist, M. Eriksen, Bone density effects on head displacement in computational models, *Ann. Biomed. Eng.* 80 (2017) 339–802.
- [33] W. Tanaka, R. Wojcik, W. Grigoryan, J. Duarte, Load path effects on flow split in computational models, *J. Biomech. Eng.* 119 (2008) 41–626.
- [34] P. Yilmaz, T. Wojcik, C. Jankowski, Friction effects on chest compression in computational models, *J. Mech. Behav. Biomed. Mater.* 112 (2023) 400–432.

Table 1: Model A credibility summary by factor and mechanism (0–4 scale; acceptance at ≥ 3).

Credibility factor	Level	Basis
Discretization error - Model A - Diastolic Coaptation Integrity	3	GCI: coaptation area 1.9%, height 1.6%, mean contact pressure 3.4% between medium and fine.
Model form - Model A - Diastolic Coaptation Integrity	3	After controlling for resolution and inputs, coaptation area bias 6.7% and mean contact pressure bias 6.8% vs bench.
Model inputs - Model A - Diastolic Coaptation Integrity	3	Thickness 0.45 ± 0.05 mm changed area by 3.8% and mean contact pressure by 6.1%; fiber angle $\pm 5^\circ$ changed height 2.2%.
Numerical solver error - Model A - Diastolic Coaptation Integrity	3	Interface mismatch $\leq 0.5\%$ of peak diastolic pressure; cycle energy defect 0.8%.
Test conditions - Model A - Diastolic Coaptation Integrity	3	Loop fit: diastolic MAPE 2.4%, systolic upstroke MAPE 4.8%; viscosity 3.4–3.5 mPa·s at 36.8 °C.
Test samples - Model A - Diastolic Coaptation Integrity	3	n=6 valves spanning 70% commissural height and 60% thickness tolerances; repeatability: height ± 0.3 mm, area ± 8 mm ² .
Discretization error - Model A - Systolic Leaflet Stress at Peak Flow	3	Peak first principal stress changed 3.6% between medium and fine; hot-spot location stable within 2 mm.
Model form - Model A - Systolic Leaflet Stress at Peak Flow	3	Magnitude bias 5.5% vs DIC-inferred stress; hot-spot within 2 mm of experiment after resolution/input controls.
Model inputs - Model A - Systolic Leaflet Stress at Peak Flow	3	$\sigma_{\max}/t$ 2.1 MPa/mm; $\sigma t = 0.05$ mm gives 0.105 MPa (~7%) stress variability.
Numerical solver error - Model A - Systolic Leaflet Stress at Peak Flow	3	Solver residual 1e6; coupling traction jump 0.4% of peak systolic traction.
Test conditions - Model A - Systolic Leaflet Stress at Peak Flow	3	Peak flow 24 ± 1 L/min; ejection time 320 ± 10 ms; impedance match within 5%.
Test samples - Model A - Systolic Leaflet Stress at Peak Flow	3	DIC-inferred peak stress 1.40 ± 0.18 MPa across n=6 valves; three runs per specimen.
Discretization error - Model A - Cycle-to-Cycle Stress Range at Commissures	3	Commissural stress range amplitude changed 4.2% under refinement; phase shift $< 1.5\%$ of cycle.
Model form - Model A - Cycle-to-Cycle Stress Range at Commissures	2	Amplitude bias 9.8% vs DIC and phase lag 6 ms suggest explicit coupling limitations in transients.
Model inputs - Model A - Cycle-to-Cycle Stress Range at Commissures	3	Dispersion ± 0.05 changed range by 0.07 MPa (11%), thickness ± 0.05 mm changed 0.04 MPa (6%).
Numerical solver	3	Time-integration damping

Table 2: Model B credibility summary by factor and mechanism (0–4 scale; acceptance at ≥ 3).

Credibility factor	Level	Basis
Discretization error - Model B - Diastolic Coaptation Integrity	4	Coaptation area 1.1%, height 0.8%, and mean contact pressure 1.9% changes from medium to fine.
Model form - Model B - Diastolic Coaptation Integrity	4	Residual bias vs bench: coaptation area 3.7%, mean contact pressure 4.1% with sharp contact enforcement.
Model inputs - Model B - Diastolic Coaptation Integrity	3	Thickness ± 0.05 mm changed area 3.2%; prestrain $\pm 2\%$ changed height 1.5%.
Numerical solver error - Model B - Diastolic Coaptation Integrity	4	Monolithic Newton to 1e8; energy defect 0.3%; contact gap < 3 μ m at closure.
Test conditions - Model B - Diastolic Coaptation Integrity	3	Pressure/flow traces met targets: diastole MAPE 2.4%, upstroke 4.8%; viscosity 3.4–3.5 mPa·s.
Test samples - Model B - Diastolic Coaptation Integrity	3	n=6 spanning 70% height and 60% thickness tolerances; coaptation repeatability ± 0.3 mm and ± 8 mm ² .
Discretization error - Model B - Systolic Leaflet Stress at Peak Flow	4	Peak first principal stress changed 2.1% between medium and fine; hot-spot stable within 1 mm.
Model form - Model B - Systolic Leaflet Stress at Peak Flow	4	Magnitude bias 3.8% vs DIC; hot-spot within 1 mm after accounting for inputs and resolution.
Model inputs - Model B - Systolic Leaflet Stress at Peak Flow	3	$\sigma_{\max}/t$ 2.0 MPa/mm; $\sigma t = 0.05$ mm gives ~0.10 MPa (6.8%) variability.
Numerical solver error - Model B - Systolic Leaflet Stress at Peak Flow	4	Residual 1e8; no coupling split; contact penetration < 2 μ m.
Test conditions - Model B - Systolic Leaflet Stress at Peak Flow	3	Peak flow 24 ± 1 L/min; ejection 320 ± 10 ms; impedance within 5%.
Test samples - Model B - Systolic Leaflet Stress at Peak Flow	3	DIC-inferred stress 1.40 ± 0.18 MPa over n=6; three runs each.
Discretization error - Model B - Cycle-to-Cycle Stress Range at Commissures	3	Stress range changed 2.8% with refinement; phase shift $< 1\%$ of cycle.
Model form - Model B - Cycle-to-Cycle Stress Range at Commissures	3	Amplitude bias 7.2% and phase lag 4 ms vs DIC across 15 cycles.
Model inputs - Model B - Cycle-to-Cycle Stress Range at Commissures	3	Dispersion ± 0.05 changed amplitude by 0.05 MPa (8%); thickness ± 0.05 mm by 0.03 MPa (5%).
Numerical solver error - Model B - Cycle-to-Cycle Stress Range at Commissures	3	Contact stabilization contributed $< 0.6\%$ damping to amplitude.
Test conditions	2	Beat-to-beat CV 3.2% and

Table 3: Model C credibility summary by factor and mechanism (0–4 scale; acceptance at ≥ 3).

Credibility factor	Level	Basis
Discretization error - Model C - Diastolic Coaptation Integrity	3	Coaptation area 2.7%, height 2.4%, mean contact pressure 4.5% changes from medium to fine.
Model form - Model C - Diastolic Coaptation Integrity	2	Bias vs bench: coaptation area 10.0% and mean contact pressure 12.1%; IB contact smear 0.2 mm.
Model inputs - Model C - Diastolic Coaptation Integrity	2	Uniform thickness and shell stiffness approximation induced 5.8% area sensitivity for ± 0.05 mm.
Numerical solver error - Model C - Diastolic Coaptation Integrity	3	CFL ~ 0.2 ; momentum non-conservation $\sim 0.1\%$; penalty gap 0.05 mm at diastole.
Test conditions - Model C - Diastolic Coaptation Integrity	3	Waveform match: diastole MAPE 2.4%, upstroke 4.8%; viscosity 3.4–3.5 mPa·s.
Test samples - Model C - Diastolic Coaptation Integrity	3	n=6 spanning major tolerances; coaptation repeatability ± 0.3 mm and ± 8 mm ² .
Discretization error - Model C - Systolic Leaflet Stress at Peak Flow	3	Peak first principal stress changed 4.2% between medium and fine; hot-spot location within 3 mm.
Model form - Model C - Systolic Leaflet Stress at Peak Flow	2	Magnitude bias 8.9% vs DIC with reduced-order turbulence and shell stiffness approximation.
Model inputs - Model C - Systolic Leaflet Stress at Peak Flow	2	$\sigma_{\max}/t$ 1.8 MPa/mm; $\sigma t = 0.05$ mm gives 0.09 MPa ($\sim 6.1\%$) variability.
Numerical solver error - Model C - Systolic Leaflet Stress at Peak Flow	3	Divergence L2 $\sim 1e4$; IB filtering attenuated peaks by 1.1% relative to unfiltered solution.
Test conditions - Model C - Systolic Leaflet Stress at Peak Flow	3	Peak flow and ejection time matched targets; impedance within 5%.
Test samples - Model C - Systolic Leaflet Stress at Peak Flow	3	DIC-inferred stress 1.40 ± 0.18 MPa over n=6; three runs per valve.
Discretization error - Model C - Cycle-to-Cycle Stress Range at Commissures	3	Stress range amplitude changed 5.1% under refinement; phase shift 9 ms.
Model form - Model C - Cycle-to-Cycle Stress Range at Commissures	2	Amplitude bias 14.3% and 9 ms phase lag due to IB smoothing and explicit coupling.
Model inputs - Model C - Cycle-to-Cycle Stress Range at Commissures	2	Dispersion ± 0.05 changed amplitude by 0.09 MPa (14%); thickness ± 0.05 mm by 0.05 MPa (8%).
Numerical solver error - Model C - Cycle-to-Cycle Stress Range at Commissures	3	Algorithmic damping yielded $\sim 1.2\%$ amplitude loss per cycle.
Test conditions - Model C -	2	Beat-to-beat CV 3.2% and ± 5 ms timing jitter constrain